

Demonstrating Spatial Haptics: A Sensory Substitution Method for Distal Object Detection Using Tactile Cues

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Figure 1: Spatial Haptics allows users to localize objects in 3D space through modulated vibrotactile feedback in handheld controllers.

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Abstract

We present Spatial Haptics, a sensory substitution method for representing locations of remote objects in 3D space via haptics. Spatial Haptics imitates auditory localization processes to enable vibrotactile localization abilities similar to those of some animal species. Two implementations of the localization method were developed, that modulate the vibration amplitude of the controllers relative to a target object in Virtual Reality. In Ear-Based Localization, vibrations are modulated based on the relative locations of the ears to the target, while in Hand-Based Localization, the amplitude is

determined based on the relative locations of the hands to the target. In this interactive demonstration, users can experience the vibrotactile localization approaches in an interactive VR mini-game. Their task is to locate the target object in a scene consisting of multiple moving objects. By experiencing spatial localization using haptics hands-on, participants can evaluate the benefits of this sensory substitution approach for detecting distal objects.

CCS Concepts

• **Human-centered computing** → **Interaction techniques**; **Virtual reality**; • **Computing methodologies** → *Virtual reality*; *Perception*.

Keywords

Haptic, Tactile, Sensory substitution, Spatial perception, Localization

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1 Introduction

Localization is the process of determining the spatial location of objects in the environment relative to oneself. This ability is achieved primarily through a combination of visual and auditory cues, and is vital for navigation, interaction, and survival [2]. For example, sound localization allows us to recognize threats and dangers such as a car rushing behind one's back, or enables us to localize an opponent in a VR shooter game that is out of our view. However, localization becomes particularly challenging in sensory-restricted environments, like VR, where traditional visual and auditory cues may be limited or overwhelming.

We present *Spatial Haptics*, a sensory substitution method that leverages the human tactile system to enhance spatial awareness and enable tactile localization. Building on related work [1, 3, 6, 7], our approach is inspired by auditory localization methods provided by the human ears and communicates the relative distance of an object through vibrations delivered to the hands. Specifically, we propose two implementations of this method. In ear-based localization (EBL), the feedback provided to each hand is determined based on the relative distance of the corresponding ear to the target. This method enables the user to move and rotate their head to acquire a better understanding of the represented location. In hand-based localization (HBL), the feedback is determined based on the relative distance of the respective hand to the target. Here, the user's hands allow them to freely explore the surrounding space to understand the target object's location. The results of our study show that both approaches enable users to detect and locate objects intuitively [8].

In this interactive demonstration, we showcase the capabilities of *Spatial Haptics* in a VR mini-game, demonstrating how the method is applicable in an accessible, consumer grade system with no modifications.

2 Spatial Haptics Implementation

Our implementation builds on the close connection between the auditory and tactile systems. It assumes that when using the palms of the hands as sensory organs for auditory-like cues, the sensory substitution would allow for performing spatial localization. In this way, we can use the dynamic ability by moving the hands to “sample more data points” than by using the ears, as well as for localizing without needing to change one's gaze. Illustrated in Figure 2, we developed two implementations of our concept:

- **Ear-Based Localization (EBL):** Vibration levels in the user's hands are determined by the relative positions of the ears to the target object. For example, when the user's right ear is directed towards the target, the user's right hand will receive vibrations at maximum amplitude while the left hand will not receive vibrations. This implementation directly translates the approach of auditory localization to haptics.
- **Hand-Based Localization (HBL):** Vibration levels in the user's hands are determined by the relative positions of the hands to the target object. For example, when the user's right hand reaches towards the target object, the right hand will receive vibrations with maximum amplitude. The vibration range is calibrated in a circle around the user with radius defined by the user's arm length. This implementation explores the possibility that one could use their hands as sensing organs for spatial location, metaphorically having ears in their hands.

We defined a vector-based algorithm that calculates a gain multiplier value between 0 and 1 based on the distance between the target object and either the user's ears or hands. This value is set using an exponential scale to mimic the way sound intensity decreases logarithmically in distance, also emphasizing the differences between distances closer to the target due to the steeper gain curve. This resonates with Weber's law, which determines that perceptual sensitivity is proportional to relative changes in stimulus intensity [4] (see Figure 3). We apply the following equations:

$$\begin{cases} m_r = (1 - \frac{|\vec{r}| - (d - a)}{2a})^2 \\ m_l = (1 - \frac{|\vec{l}| - (d - a)}{2a})^2 \end{cases}$$

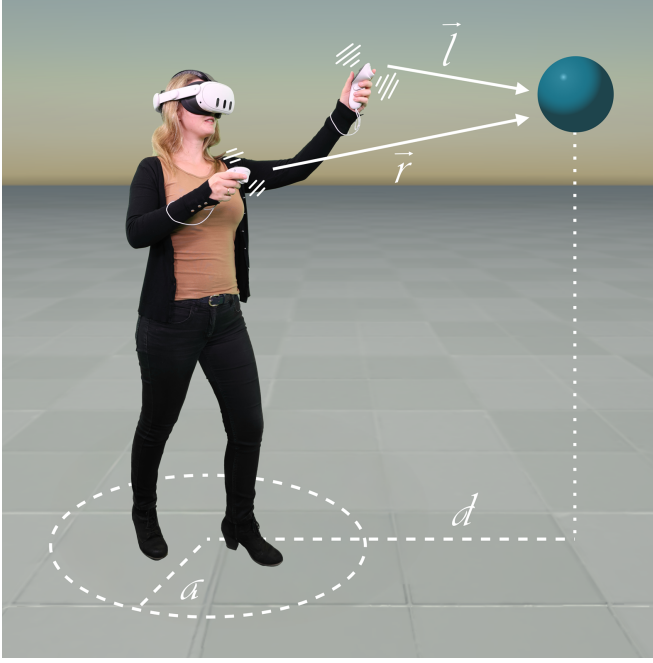
where a is the user's arm length for HBL and head radius for EBL, and d is the distance to the target object. This vector-based algorithm approach enables real-time calculation, requiring lower computational power compared to complex spatial audio algorithms, making it an efficient and applicable way for conveying location of virtual objects using vibrations.

3 Demonstration of Spatial Haptics

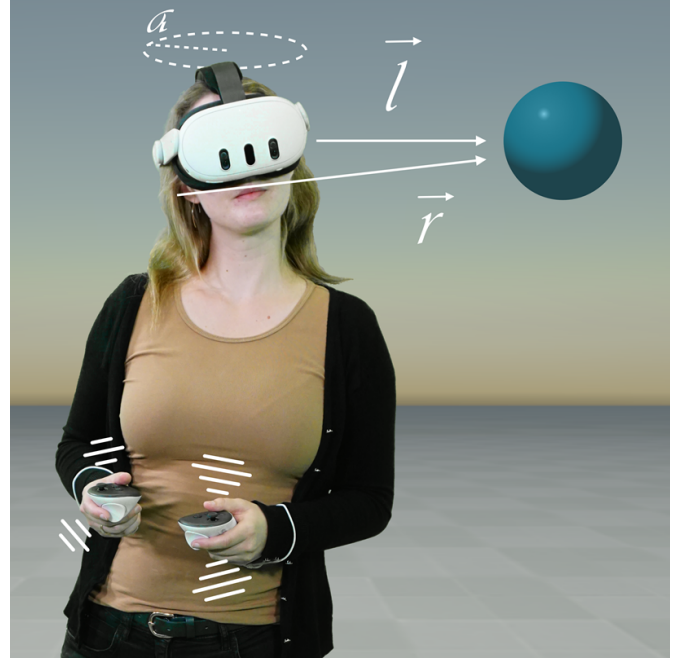
Our interactive demonstration allows users to experience sensory substitution to localize objects in 3D space via haptics.

3.1 Method

In our demonstration, we chose to focus on the hand-based localization method. The method is unique in its approach, since it uses the hands as sensory objects for spatial perception, acting as “ears in



(a) Hand-based Localization Method (HBL)



(b) Ear-based Localization Method (EBL)

Figure 2: Spatial Haptic Localization Methods: (a) **Hand-based Localization Method:** Vibration intensity varies according to the distance vectors ($\vec{l}$ and $\vec{r}$) from the controllers to the object in space. The user is positioned within a circle of radius r , and d denotes the distance to the spherical object; (b) **Ear-based Localization Method:** Vibration intensity varies according to the distance vectors ($\vec{l}$ and $\vec{r}$) from the ear positions to the object in space. Each vector respectively determines the intensity of the left and right controllers.

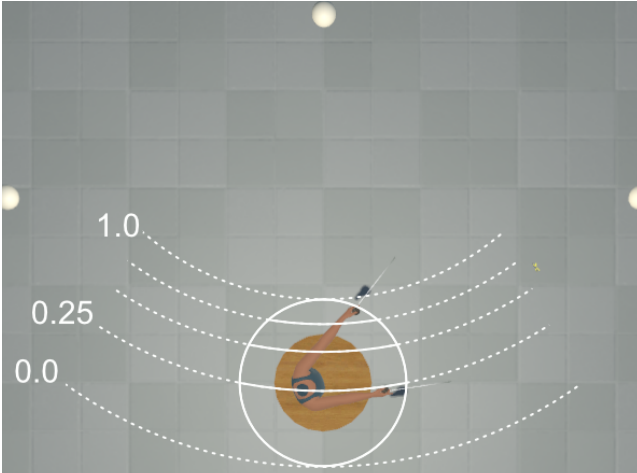


Figure 3: Algorithm visualization of the vibration feedback mechanism: Vibration is “emitted” from the target object (at the top of the image). Intensity decreases with distance exponentially, with the maximum, 1.0 value when a controller is closest to the object and decreases down to 0 at the furthest distance from the target within the user’s reach, providing spatial feedback to the user based on proximity to the target.

the hands”. It was also interesting to see users of the method relying on comparative use of the two hands. This method has advantages as it enables users to move their hands and “sample more points of data”. This advantage was apparent in the results of our study [8], which showed that front-back confusion was almost completely mitigated when this method was used.

3.2 Procedure

Our experience starts with a brief introduction on-boarding into the concept of tactile localization through sensory substitution, followed by a short mini-game that tests participants’ skill in localizing target objects. An overview of the procedure is shown in Figure 4.

Upon starting the experience, a presenter will explain the concept of Spatial Haptics, assist the participant in mounting the head-mounted display, and hand them the controllers. To familiarize themselves with the setup, participants first experience a short scenario in which the location of a single target sphere is communicated using vibrations. During this scripted introduction, the target object will demonstrate different movements in space, allowing the participants to familiarize with the method’s mechanisms. After a fixed duration, the experience will continue with the mini-game.

The VR mini-game consists of several rounds in which participants are tasked with localizing and selecting a target object. In each round, participants are surrounded by several moving objects of

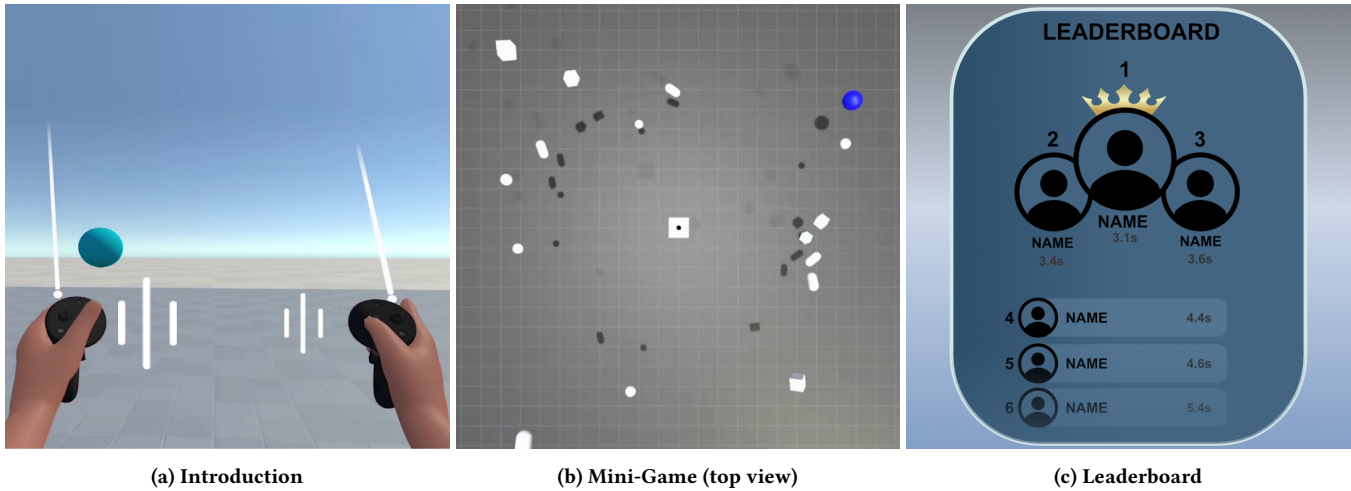


Figure 4: Our interactive demonstration consists of three elements: (a) Participants will first receive an introduction into the concept of Spatial Haptics; (b) In a mini-game, participants are tasked to localize a target object with or without spatial localization; (c) A leaderboard will show live scores such that participants can compete with each other.

varying shapes. At preset times, a target object, differently colored from all other objects, appears in the scene outside of the user’s sight. A timer appears, and participants are asked to locate the target as fast as possible (which can appear at any location behind, below, or above them). In each trial, participants are either guided by the vibration in their hands, responsive to the location of the target, or receive no tactile guidance, using vision alone. This allows participants to experience the added value of tactile localization, specifically for cases where the target is outside the participant’s field of view. To confirm their selection, participants must aim a yellow crosshair at the center of their view at the target, and press one of the thumb buttons. Upon selection, the target disappears. Each round has a limited duration. The round continues until the target is located or until time runs out. Should the participant be unable to select the target object within the given time, the end of round will be indicated, and the maximum time recorded as the trial’s duration.

Upon completing the mini-game, participants will be shown their performance for rounds with and without tactile localization. A leaderboard will indicate their placement compared to other people who participated in the demonstration. The leaderboard and a teaser video will be displayed on screen for spectators not currently participating in the demo.

3.3 Apparatus

Our interactive demonstration was developed using Unity3D (version 2022.03.32f1) and deployed on the Quest 3 VR Headset. The hardware was chosen for its portability, and because its controllers use voice coil actuators that support a wide band of frequencies and complex signals. Vibrations are provided using a constant 125Hz frequency signal as previous research indicated that for hand-transmitted vibration, specifically grip, sensitivity peaks around this frequency [5]. By modeling a function mapping the parametric

frequency value (0-1) in the Meta Haptics SDK to the physical frequency (Hz), we determined the value for the frequency parameter to be 0.485.

4 Conclusion

In our interactive demonstration, we present Spatial Haptics, a sensory substitution method for representing locations of remote objects in 3D space via haptics. Through a VR mini-game, participants will be able to experience the added value of tactile localization. By presenting participants with their performance results with tactile localization as compared to without, our work highlights the added value of the approach presented. By experiencing tactile localization hands-on, participants can understand the intuitiveness of our sensory substitution approach. We hope to engage in meaningful discussions around the topic.

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